Node Lifecycle Hypothesis and the Crystallisation of 3D Space in CRF

Version 3: V20.3 Simulation Results

d_s = 3.031 ± 0.076*acrossthreeindependentruns.Phasegappositiveinallruns.Transferraterevealslossycrystallisati*

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March 2026

Abstract

CRF Hybrid V20.3 provides the most stable simulation evidence to date for the Node Lifecycle Hypothesis.

Three independent runs (N=500, 400 sweeps each) give:

$$\bar{d}_s = 3.031 \pm 0.076 \qquad \text{(target 3.0 from SO(3) proof)}$$

The standard deviation has dropped from 0.233 (V20.2) to 0.076, indicating consistent convergence.

The phase test — comparing crystallised (Ω-dominant) nodes against young (IDV-dominant) nodes — is positive in all three runs:

$$\Delta d_s = d_s^Y - d_s^X = 0.980 \pm 0.240 \quad \text{(positive in 3/3 runs)}$$

The per-collapse transfer rate measurement reveals a new finding: crystallisation is not a 1:1 exchange of IDV for Ω. The measured ratio $\Delta\mathrm{IDV}/(-\Delta\Omega) = -2.212 \pm 0.015$ means that each R-event reduces IDV by approximately twice what it adds to Ω. This is a *lossy* process: not all of the internal dimensional volume converts to mass — some possibility is simply relinquished.

Gravity $R^2 = 0.9071 \pm 0.0010$, the most stable value across all simulator versions, confirming that the wave and lifecycle extensions do not disturb the emergent gravitational law.

Five sub-hypotheses (HN-01 through HN-05) are updated with V20.3 evidence. HN-04 (Dimensional Hysteresis) is confirmed as a corollary of the SO(3) proof. The transfer rate finding motivates a revision of HN-01.

Contents

1 V20.3: Three Fixes and What They Revealed

V20.3 introduced three targeted improvements over V20.2:

V20.3 Changes

Fix 1 — Per-collapse transfer measurement: Instead of tracking global $\overline{V^{\text{int}}}$ and $\overline{\Omega}$ per sweep, V20.3 records IDV and Ω immediately before and after each R-event, giving a direct measurement of the transfer at the moment it occurs.
Fix 2 — Absolute phase threshold: Groups X (crystallised) and Y (young) are defined by absolute Ω percentiles (70th and 30th), not the top/bottom 20%. This ensures the groups represent distinct lifecycle stages.
Fix 3 — 8 trials for subgroup d_s: The subgroup estimator uses 8 independent walk sets (was 3), giving ∼800 effective walkers per group and much lower variance in the phase gap measurement.

2 Core Results

2.1 Spectral Dimension

\$d$_s$\$ $= 3.0310.076(3 runs)$

Run	d_s (late mean)	Gravity R^2
1	3.056	0.9057
2	3.110	0.9076
3	2.928	0.9081
Mean ± std	$\mathbf{3.031 \pm 0.076}$	$\mathbf{0.9071 \pm 0.0010}$

Comparison across simulator versions:

Version	d_s	Gravity R^2	Phase gap
V16 (canonical base)	3.008	0.906	—
V18 (IDV paper)	3.335	0.908	wrong direction
V20 (Omega-metric)	3.238	0.900	+1.58 (1 run)
V20.2 (continuous d_s)	2.976 ± 0.233	0.904	+0.88 (3/4 runs)
V20.3 (this)	$\mathbf{3.031 \pm 0.076}$	**0.907**	**+0.98 (3/3)**

2.2 Phase Test: Node Lifecycle Confirmed

Phase Gap Positive in All Runs

Nodes split by Ω level:

- Group X: $\Omega \geq 70$th percentile (crystallised)
- Group Y: $\Omega \leq 30$th percentile (young)

$$d_s^X = 3.08 \pm 0.22 \qquad d_s^Y = 4.06 \pm 0.48 \qquad \Delta = 0.980 \pm 0.240$$

All three runs: $\Delta > 0$.
Interpretation: crystallised nodes (Ω-dominant) consistently reside closer to $d_s = 3$ than young nodes (IDV-dominant). This is the central prediction of the Node Lifecycle Hypothesis, now supported in all simulation runs.

2.3 Transfer Rate: A New Finding

Finding 1 (Lossy Crystallisation)**.** The per-collapse transfer rate, measured directly at each R-event:

$$r = \frac{\Delta\text{IDV}}{-\Delta\Omega} = \frac{\text{IDV}_{\text{after}} - \text{IDV}_{\text{before}}}{-(\Omega_{\text{after}} - \Omega_{\text{before}})} = -2.212 \pm 0.015$$

(measured over 4663 collapse events per run).

Each R-event increases Ω by $\kappa = 0.15$ (fixed, from mass $= 1 + \kappa \cdot N_{\text{eff}}$). The measured ratio means IDV decreases by $\approx 2.2 \times 0.15 = 0.33$ per event — more than twice the Ω gain.

Interpretation: crystallisation is not a 1:1 exchange. For every unit of Ω (crystallised history) gained, approximately 2.2 units of IDV (accessible states) are lost. Some of the internal dimensional volume is simply relinquished — not converted to mass, but abandoned.

This is the CRF analogue of a thermodynamic efficiency $< 100\%$: the process of becoming actual costs more possibility than the actuality it creates.

3 Sub-Hypothesis Updates

3.1 HN-01 Revised: Lossy Phase Transition

The original HN-01 proposed $F_i = H/(\Omega + \varepsilon_0)$ decreases at crystallisation (correct), but implicitly assumed a 1:1 IDV–Ω exchange.

The transfer rate finding revises this:

$$\Delta\text{IDV} \approx -\beta \cdot \Delta\Omega \qquad \beta \approx 2.2 \text{ (measured)}$$

The free ratio at crystallisation satisfies $\Delta F_i \approx -(1+\beta)\Delta\Omega/(\Omega + \varepsilon_0)^2$, falling faster than a 1:1 model predicts. The factor β is an empirical constant of V20.3 awaiting derivation from the δ-chain.

3.2 HN-04: Confirmed as Corollary

Corollary 1 (of SO(3) Proof)**.** Dimensional hysteresis — once $d_s \to 3$, perturbation cannot restore a stable high-D state — follows directly from Theorem 1 of CRF_SO3_Proof. Since $n = 3$ is the unique δ-stable dimension, no other stable attractor exists to receive a perturbation.

V20.3 d_s oscillations (range 1.2–4.5 within runs) are consistent with this: the system fluctuates but the *late mean* remains near 3 in all runs.

4 Complete Status Table

Code	**Statement**	**Status**	**Evidence**
Core	IDV-phase $\to$ Ω-phase	Hypothesis	V20.3 phase gap 3/3
Core	$d_s \to 3$ at crystallisation	**Confirmed**	3.031 ± 0.076
Core	Phase gap $\Delta > 0$	**Confirmed**	3/3 runs
New	Lossy crystallisation $\beta \approx 2.2$	**Finding**	4663 events/run
HN-01	F_i drops, lossy	Hypothesis	V20.3 F_i trend
HN-02	Aging at $H = \varepsilon_0$	Hypothesis	HDynamics
HN-03	Collective Ω excess	Hypothesis	CRF Synchronicity
HN-04	Hysteresis (no back-path)	**Corollary**	SO(3) Proof
HN-05	Causal geometry mapping	Hypothesis	$t = \lvert R_{\text{events}} \rvert$
Grav.	$R^2 = 0.907 \pm 0.001$	**Confirmed**	3 runs
Spec.	7–9 Ψ modes	**Confirmed**	V20.3 histogram

5 What Remains Open

Open: Derive $\beta \approx 2.2$ from δ-chain

The transfer ratio $\beta = \Delta\mathrm{IDV}/(-\Delta\Omega) \approx 2.2$ is empirical. It should be derivable from the IDV formula and the Born rule resolution step. Specifically: when P undergoes Dirichlet sampling, how much does $H(P_i)$ decrease on average? The answer is a function of D_0, ε_0, and k. This is the next formal derivation target.

Open: P0 experiment

$G_{\mathrm{sim}}/\ln 2 = 0.196\text{-}0.205$ across V20.3 runs. The gap to 1.0 requires measuring D_0 in SI units through the P0 experiment ($\tau_1 < \tau_2$ with SNSPD detectors). This remains the gate to Level 7/8 maturity.

Crystallisation does not conserve possibility.
For every unit of mass gained,
more than two units of "could be" are released.
The universe pays more than it receives —
and what it receives is what we call matter.